2021-10585 김찬홍 13주차

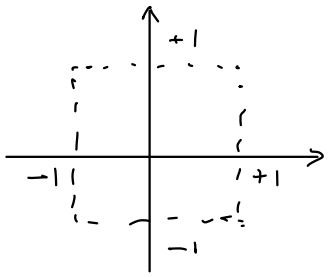
1번. construct set C in plane

that is ① convex

② symmetric about ⓞ

③ area 4

④ contains no nonzero integral point.



$C = \{(x,y) \in \mathbb{R}^2 : |x|<1,\ |y|<1\}$

$= (-1,1) \times (-1,1)$

- 넓이 = 가로 세로 $2 \cdot 2 = 4$

- $(x,y) \in C$ then $|x|, |-x|, |y|, |-y| < 1$ (절댓값 대칭)

$(-x,-y) \in C$

- 정수점 in C $\iff$ $(x,y) \in \mathbb{R}^2$ $\iff$ $x=y=0$

$-1<x,y<1$ (원점)

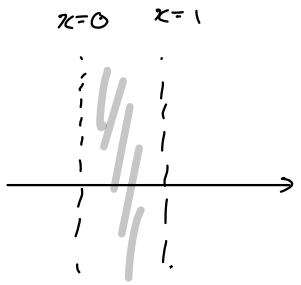
- 정사각형 내부의 두 점 사이를 이으면 $\Rightarrow$ convex

선분이 정사각형 안에 포함됨

2번. set C in plane ① convex

② infinite area

③ no integral point.



$C = \{(x,y) \in \mathbb{R}^2 : 0<x<1\}$

- x 좌표가 0, 1 사이 $\Rightarrow$ 선분 위 x 좌표도 0과 1 사이

- area(C) = 밑변 × 높이 = ∞

$1 \times \infty$

- $(m,n) \in \mathbb{Z}^2$ $\iff$ $0<m<1$ $\rightarrow\leftarrow$

증명. if ① $a \not\equiv 0 \pmod p$ $\Rightarrow$ $\exists x, y \in \mathbb{Z}$

② p prime

③ any positive integer n

s.t ① $x \equiv ay \pmod p$

② $0 < x \le n$

③ $|y| \le \frac{p}{n}$

- $\Lambda = \{(x,y) \in \mathbb{Z}^2 : x \equiv ay \pmod p\}$

$= \{m(p,0) + n(a,1) : m, n \in \mathbb{Z}\}$

∵ $x - ay = pk$ $(\exists k)$

$\Rightarrow x = pk + ay$

$y = y$

$\Rightarrow k \cdot (p,0) + y(a,1)$

- $\det \Lambda = \det \begin{bmatrix} p & 0 \\ a & 1 \end{bmatrix} = p$

∵ $x = mp + na \Rightarrow x \equiv na \pmod p$

$y = n$

- $C_\varepsilon = \{(x,y) \in \mathbb{R}^2 : |x| < n+\varepsilon,\ |y| < \frac{p}{n} + \varepsilon\}$ $(\varepsilon > 0)$

$\text{area}(C_\varepsilon) = 2(n+\varepsilon) \cdot 2\left(\frac{p}{n} + \varepsilon\right) > 4p = 4\det(\Lambda)$

because $\varepsilon > 0$

- by Minkowski C_ε에는 Θ이 아닌 격자점이 존재한다.

i.e $\exists (x,y) \neq (0,0) \in \mathbb{Z}^2$ s.t ① $x \equiv ay \pmod p$

② $|x| < n+\varepsilon$, $|y| < \frac{p}{n} + \varepsilon$

with $\varepsilon \to 0$ $n \le |x|$ $|y| \le \frac{p}{n}$

- if $x = 0$ then $0 \equiv ay \pmod p$ ∵ $a \not\equiv 0 \pmod p$

$a^{-1} \cdot 0 \equiv a^{-1}ay$ $(\because)$

$0 \equiv y$ $(\because)$

however $|y| \le \frac{p}{n}$, $|y| < p$ if $n \ge 2$ $\Rightarrow y = 0$ $\rightarrow\leftarrow$

- $x \equiv ay \pmod p$ then $-x \equiv a(-y) \pmod p$ 따라서 $(-x,-y)$ 또한 성립.

- $n = 1$ then $aa^{-1} \equiv 1 \pmod p$ then $0 < x \le 1$, $|y| \le p$.

(!! y, !! x)

14. any lattice Λ in plane

contains nonzero (x,y) s.t $|xy| \le \frac{1}{2} d(\Lambda)$

- by Minkowski

: Λ에 대해 ① 원점 대칭

② convex 인 C 가 $\text{area}(C) > 4d(\Lambda)$

$\Rightarrow$ Λ에는 nonzero 격자점이 있다.

- $C_\varepsilon = \{(x,y) \in \mathbb{R}^2 : \frac{|x|}{A} + \frac{|y|}{B\varepsilon} < 1\}$ (타원)

with ① $A > 0$

② $AB\varepsilon = 2d(\Lambda) + \varepsilon$

$\longrightarrow$ { 원점 대칭, convex

$\text{넓이} = 2AB\varepsilon$ Minkowski!

$= 2(2d(\Lambda) + \varepsilon)$

$= 4d(\Lambda) + 2\varepsilon$

- C_ε에는 nonzero 격자점 있음 -

i.e $\frac{|x|}{A} + \frac{|y|}{B\varepsilon} < 1$ and $(x,y) \neq (0,0)$

- $u = \frac{|x|}{A}$, $v = \frac{|y|}{Bc}$ $u+v < 1$ $\Rightarrow$ $uv \le \left(\frac{u+v}{2}\right)^2 < \frac{1}{4}$

$\Rightarrow$ $|xy| < AB\varepsilon/4$ $\Rightarrow$ $\varepsilon \to 0$

$= \frac{1}{2} d(\Lambda) + \frac{1}{3}$ $d(\Lambda) \frac{1}{2} \le |xy|$

- $|xy| = \frac{1}{2} d(\Lambda)$ 일 때 $\frac{|x|}{A} = \frac{|y|}{B} = \frac{1}{2}$ $\Rightarrow$ $|x| \neq \frac{A}{2}$ 가

$|x| = \frac{A}{2}$ 일지라도 A를 조금 바꾸면

lattice generator A that can replace the matrix used in Step 1 of pf of Lagrange four square thm. while still satisfying necessary cond for proof.

- choose $A' = \begin{bmatrix} p & 0 & \nu & s \\ & p & -s & \nu \\ & & 1 & \\ & & & 1 \end{bmatrix}$

- choose integers s.t $\nu^2 + s^2 \equiv -1 \pmod p$

- lattice $\Lambda = A'\mathbb{Z}^4$

- 임의의 $\vec{x} \in \Lambda$에 대해 $\vec{t} = (t_1, t_2, t_3, t_4)$가 존재.

$$\Rightarrow \vec{x} = A'\vec{t} = \begin{bmatrix} pt_1 + \nu t_3 + s t_4 \\ pt_2 - s t_3 + \nu t_4 \\ t_3 \\ t_4 \end{bmatrix}$$

- $x_1^2 + x_2^2 + x_3^2 + x_4^2 = (pt_1 + \nu t_3 + s t_4)^2 + (pt_2 - s t_3 + \nu t_4)^2 + t_3^2 + t_4^2$

$\pmod p \equiv \nu^2 t_3^2 + 2\nu s t_3 t_4 + s^2 t_4^2 + s^2 t_3^2 - 2\nu s t_3 t_4 + \nu^2 t_4^2 + t_3^2 + t_4^2$

$\equiv (\nu^2 + s^2 + 1)(t_3^2 + t_4^2)$

$\equiv 0$

$\det A' = p \cdot p \cdot 1 \cdot 1 = p^2$ (부피가 p^2배)

① $\forall \vec{x} \in A'\mathbb{Z}^4$ 에 대해 $x_1^2 + x_2^2 + x_3^2 + x_4^2 \equiv 0 \pmod p$

② $\det A' = p^2$

3ex. infinite conti fractions

(a) $\langle 2,2,2,2,\ldots\rangle$

$x = 2 + \frac{1}{x} \Rightarrow$ since $x>0$

$x^2 = 2x+1$ $\quad x = 1+\sqrt{2}$

$x^2 - 2x - 1 = 0$

$x = 1 \pm \sqrt{2}$

(b) $\langle 1,2,1,2,1,2\ldots\rangle$

$x = 1 + \frac{1}{2+\frac{1}{x}}$ $\quad x>0$

+ Cecafore $\quad \frac{1+\sqrt{3}}{2} = x$

$x-1 = \frac{x}{2x+1}$

$(x-1)(2x+1) = x$

$x = \frac{1 \pm \sqrt{3}}{2}$

(c) $\langle 2,1,2,1\ldots\rangle$

$x = 2 + \frac{1}{1+\frac{1}{x}}$ $\quad x>0$

+ Cecafore

$(x-2) = \frac{x}{(x+1)}$ $\quad x = 1+\sqrt{3}$

$(x-2)(x+1) = x$

$x = 1 \pm \sqrt{3}$

(d) $\langle 1,3,1,2,1,2\ldots\rangle$ $\quad y$

$x = 1 + \frac{1}{3+\frac{1}{y}}$

$= 1 + \frac{y}{3y+1}$ $\quad$ with $y = \frac{1+\sqrt{3}}{2}$

$= 1 + \frac{1+\sqrt{3}}{5+3\sqrt{3}}$

$= 1 + (2-\sqrt{3}) = 3-\sqrt{3}$

4ex. $(n \geq 1)$ $\quad k_n / k_{n-1} = \langle a_n, a_{n-1}, \ldots, a_2, a_1\rangle$

similar continued fraction expression for h_n/h_{n-1} $(a_0 > 0)$

Ex:

$h_n = a_n h_{n-1} + h_{n-2}$ $\quad (h_{-2}, k_{-2}) = (0,1)$

$k_n = a_n k_{n-1} + k_{n-2}$ $\quad (h_{-1}, k_{-1}) = (1,0)$

$(h_0, k_0) = (a_0, 1)$

- $k_n = a_n k_{n-1} + k_{n-2}$

$\langle a_n, a_{n-1}, \ldots, a_2, a_1\rangle$

$\frac{k_n}{k_{n-1}} = a_n + \frac{1}{k_{n-1}/k_{n-2}} = a_n + \frac{1}{a_{n-1} + \frac{1}{k_{n-2}/k_{n-3}}} = \cdots = a_n + \frac{1}{a_{n-1} + \frac{1}{\ddots + \frac{1}{a_1}}}$

- $h_n = a_n h_{n-1} + h_{n-2}$.

with $\frac{h_1}{h_0} = \frac{a_1 a_0 + 1}{a_0} = a_1 + \frac{1}{a_0}$

$\frac{h_n}{h_{n-1}} = a_n + \frac{1}{k_{n-1}/k_{n-2}} = a_n + \frac{1}{a_{n-1} + \frac{1}{\ddots + \frac{1}{h_1/h_0}}}$ $\quad a_1 + \frac{1}{a_0} = \langle a_n, a_{n-1}, \ldots, a_1, a_0\rangle$

5번. $u_0/u_1 = \langle a_0, a_1, \ldots, a_n \rangle$

: rational number in lower terms

$\Rightarrow$ if $0 \le i < n$ then $\left| r_i - \frac{u_0}{u_1} \right| \le \frac{1}{k_i k_{i+1}}$

equality iff $i = n-1$

pf.

- $\frac{u_0}{u_1} = r_n = \frac{h_n}{k_n}$

- $h_i k_{i-1} - h_{i-1} k_i = (-1)^{i-1}$

  $|h_i k_{i-1} - h_{i-1} k_i| = 1$

- $r_i = h_i/k_i$, $r_n = h_n/k_n$ $\Rightarrow$ $|r_i - r_n| = \left| \frac{h_i}{k_i} - \frac{h_n}{k_n} \right| = \frac{|h_i k_n - h_n k_i|}{k_i k_n} \le \frac{1}{k_i k_n}$

  $\le \frac{1}{k_i k_{i+1}}$

$\Rightarrow \left| r_i - \frac{u_0}{u_1} \right| \le \frac{1}{k_i k_{i+1}}$

- 등호 성립 $\Rightarrow$ $k_n = k_{i+1}$

  $a_{i+1} k_j = a_j k_{j-1} + k_{j-2}$ 이므로 $n = i+1$ (부등이 항 無)

  ($a_j \ge 1$)

- $i = n-1 \Rightarrow r_{n-1} = r_i \Rightarrow \left| r_i - \frac{u_0}{u_1} \right| = \frac{1}{k_i k_{i+1}}$

  $r_n = u_0/u_1$

  $|r_{n-1} - r_n| = \frac{1}{k_{n-1} k_n}$

Goal. ① prime $p \equiv 1 \pmod 4$

② integer $u^2 \equiv -1 \pmod p$

③ $\frac{u}{p} = \langle a_0, a_1, \dots, a_n \rangle$

$\frac{h_j}{k_j} = \langle a_0, a_1, \dots, a_j \rangle$

④ i는 $k_i \le \sqrt{p}$ 인 가장 큰 정수

Claim $i < n$

- $h_n / k_n = u/p$ but $p > \sqrt{p}$ $\Rightarrow$ $k_i \le \sqrt{p} < k_{i+1}$
- $k_n = p$ $k_n = p > \sqrt{p} \ge k_i$

$n > i$

Claim $\left| \frac{h_i}{k_i} - \frac{u}{p} \right| < \frac{1}{k_i \sqrt{p}}$, $|h_i p - u k_i| < \sqrt{p}$

유리수 $u/p = \langle a_0, a_1, \dots, a_n \rangle$

$\Rightarrow$ $0 \le j < n$ 인 $\left| \frac{h_j}{k_j} - \frac{u}{p} \right| \le \frac{1}{k_j k_{j+1}}$ $\Rightarrow$ $\left| \frac{h_i p - u k_i}{k_i p} \right| < \frac{1}{k_i \sqrt{p}}$

$\left| \frac{h_i}{k_i} - \frac{u}{p} \right| \le \frac{1}{k_i k_{i+1}} < \frac{1}{k_i \sqrt{p}}$ $|h_i p - u k_i| < \frac{k_i p}{k_i \sqrt{p}} = \sqrt{p}$

Claim $0 < x^2 + y^2 < 2p$

let $(x, y) = (k_i, h_i p - u k_i)$ $\Rightarrow$ $x^2 \le p$ $\Rightarrow$ $x^2 + y^2 < 2p$

$\le \sqrt{p}$ $< \sqrt{p}$ $y^2 < p$

meanwhile $x = k_i > 0$ therefore $x^2 + y^2 > 0$

$0 < x^2 + y^2 < 2p$

$\Rightarrow$ $x^2 + y^2 = p$

Claim $x^2 + y^2 \equiv 0 \pmod p$

$x = k_i$

$y = h_i p - u k_i \equiv -u k_i \pmod p$

$\equiv -u x$

$\Rightarrow$ $y^2 \equiv u^2 x^2$

$x^2 + y^2 \equiv (1 + u^2) x^2$

$\equiv (1 - 1) x^2 \equiv 0$

two distinct inf simple continued fractions converge to diff values

$\alpha = \langle a_0, a_1, \ldots \rangle \qquad a_n \neq b_n$

$\beta = \langle b_0, b_1, \ldots \rangle$

if $\alpha = \beta$ $\quad \alpha = a_0 + \dfrac{1}{\langle a_1, a_2 \ldots \rangle}$ then $\langle a_1, a_2, \ldots \rangle > 1$

$\downarrow$

between 0 and 1

$\Rightarrow \quad a_0 < \alpha < a_0 + 1 \qquad \Rightarrow \quad a_0 = b_0$

$a_0 = \lfloor \alpha \rfloor = \lfloor \beta \rfloor = b_0$

$$\frac{1}{\langle a_1, a_2, \ldots \rangle} = \alpha - a_0 = \beta - b_0 = \frac{1}{\langle b_1, b_2 \ldots \rangle}$$

$\langle a_1, a_2 \ldots \rangle = \langle b_1, b_2 \ldots \rangle \quad \Rightarrow \quad a_1 = b_1$

$\Rightarrow \quad a_2 = b_2$

$\ldots$